

# Scientific Paper Template

## Design Paper for Oral Exam / Design Brief Summary

*Maximum length: 10 pages + images / diagrams*

*Use this document as a starting structure. You may integrate, reorganise or expand sections, but keep the final paper concise, evidence-based and supported by diagrams and references.*

### A Very Holy Mess

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### Abstract

*Write 150–200 words. State the problem, the design objective, the proposed game/experience, the method, and the expected contribution.*

#### Abstract text

A Very Holy Mess is a cooperative hybrid board game with a companion app, designed for the *Pinacoteca Nazionale di Bologna*. The project addresses the difficulty of engaging young visitors with a large and traditionally displayed collection, where artworks can easily be perceived passively or become overwhelming. The design objective is to transform the museum visit into an active, playful, and repeatable experience that encourages close observation, interpretation, and emotional involvement with the artworks. Starting from Vitale da Bologna's *San Giorgio e il drago*, players become Saint George's guardian angel and must help him defeat the dragon by completing quests across the museum. Through physical cards, a divided game board, timed movement, challenge decks, and app-based interactions, players investigate paintings, ask questions, and solve mini-games connected to the collection. The project was developed through museum benchmarking, collection and spatial analysis, user persona creation, CoDesign, and the PACT framework. Its main contribution is a flexible hybrid format that can be played both inside the museum through AR interactions and at home through artwork cards and QR-based digital content.

#### Keywords (5–7)

hybrid board game; museum engagement; cultural heritage; art education; augmented reality; co-design; Pinacoteca Nazionale di Bologna

## 1. Introduction and Design Problem

*Introduce the museum/context, why the project is needed, and which design problem your project addresses. Avoid beginning with technical features only.*

### 1.1 Context of the project

The *Pinacoteca Nazionale di Bologna* is a museum with a rich and extensive collection of artworks, many of which belong to religious, historical, and narrative traditions. These paintings are culturally valuable, but they can also be difficult for young or non-specialist visitors to approach. Without specific mediation, visitors may struggle to understand what they are looking at, which details are important, and how the artworks relate to each other.

The project starts from this visitor-experience problem. In a traditional museum visit, the audience is often placed in the role of passive observer: they move through the rooms, look at the artworks, read labels, and continue to the next painting. This format can make the visit feel overwhelming, especially in a museum with many visually complex works. As a result, younger visitors may lose attention quickly or perceive the artworks as distant from their own interests and ways of learning.

The design problem addressed by the project is therefore how to make young visitors interact more deeply with the museum collection without reducing the artworks to simple decoration or entertainment. The goal is to create an experience that encourages attention, interpretation, movement, and curiosity, while remaining accessible both inside the museum and beyond the physical visit.

## 1.2 Main pain points / problems the project aims to solve

During the analysis of the museum context, several pain points emerged that the project aims to address through the design of a hybrid board game and companion app:

- The museum collection is rich and extensive, but the large number of artworks can make the visit feel overwhelming, especially for young or non-specialist visitors.
- Visitors may observe paintings quickly and superficially, without noticing narrative details, symbols, gestures, characters, or connections between artworks.
- Many artworks represent religious or historical subjects that can feel distant from contemporary young audiences if they are not mediated through an accessible narrative.
- The museum lacks a strong digital extension of the visit, limiting the possibility of remote engagement, pre-visit preparation, and post-visit continuity.
- The visit does not naturally encourage return visits.

| Pain point / problem                                                                    | Who is affected?  | Why does it matter for the design?                                                                                                                                                     |
|-----------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The museum contains many artworks, which can make the visit feel visually overwhelming. | Audience          | The game must divide the museum into clear zones and give players specific objectives, reducing cognitive overload.                                                                    |
| Visitors may look quickly and miss important details.                                   | Audience / Museum | The game must slow down observation.                                                                                                                                                   |
| Religious and historical subjects may feel distant from young audiences.                | Audience          | The game must create a playful narrative entry point, making the player feel emotionally connected to the protagonist of the narrative and making the artworks emotionally accessible. |
| The museum has limited digital extension outside the physical visit.                    | Museum            | The game must allow the experience to continue at home.                                                                                                                                |
| The visit often lacks a reason to be repeated.                                          | Museum            | The game must support re-playability through randomised tasks and at-home play.                                                                                                        |

## 2. Related Work, Benchmarks and Existing Solutions

Discuss published work, comparable projects, technologies, museum games, apps, photography learning tools or digital heritage experiences. Explain what they solve and what gap remains.

### 2.1 Existing references, comparable projects and technologies

Since the main idea was to make a hybrid board game with a companion app, the main references were *Murder Party Pocket* and *Chronicles of Crime*. Both games combine physical cards with digital interaction, showing how an app can guide narrative progression without replacing the social and physical dimension of tabletop play.

*Murder Party Pocket* inspired the use of the app as a game master: a tool that guides the investigation, manages the flow of the story, and supports the use of physical cards. *Chronicles of Crime* was the main reference for QR-based interaction, where scanning cards unlocks clues, characters, locations, and narrative consequences. This directly influenced the idea of Artwork Cards, which allows players to interact with paintings through the companion app.

As secondary references, we considered *The Museum Mystery* by Useeum and *Mostri agli Uffizi* by the Gallerie degli Uffizi. These projects show how a museum visit can become a playful exploration based on clues, hidden details, and narrative discovery. However, they remain closer to museum activities than to structured hybrid board games.

| Reference / project / technology | What it does well                                                                              | Gap for your project                                                                                |
|----------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Murder Party Pocket              | App-guided investigation, physical cards, narrative progression, companion app as game master. | It's not re-playable and doesn't improve the player's general knowledge on cultural heritage.       |
| Chronicles of Crime              | QR-code interaction, cooperative investigation, strong physical-digital integration.           | It's only playable at home and doesn't improve the player's general knowledge on cultural heritage. |
| The Museum Mystery               | Museum exploration, child-friendly investigation, playful movement through physical spaces.    | It is closer to a digital scavenger hunt than to a hybrid board game.                               |
| Mostri agli Uffizi               | Playful observation of hidden details and creatures in artworks.                               | It is a guided museum itinerary, not a replayable hybrid game with cards and app interaction.       |

## 2.2 Gap statement: what existing solutions do not solve

The analysis of these references revealed a clear gap: hybrid investigation board games successfully combine physical cards, companion apps, QR codes, and narrative progression, but they are not designed for cultural heritage or direct artwork interpretation. At the same time, museum activities can make artworks more accessible through stories, discovery, and playful observation, but they rarely offer the structure, re-playability, and phygital integration of modern hybrid board games. The gap addressed by the project is the creation of a re-playable hybrid board game that uses these hybrid games' mechanics not to solve a fictional crime, but to make players observe, question, and emotionally connect with real artworks from the *Pinacoteca Nazionale di Bologna*.

## 3. Method and Design Process

*Describe how you designed the project: co-design tool, classworks, concept design, game loop, mechanics, user flow, swimlane, user journey, storyboard, prototype sketches. This section makes the paper scientific, not only descriptive.*

### 3.1 Design method used

The game was developed using a human-centred approach, combining museum analysis, game design, the use of CoDesign, and prototyping tools. The starting point was the observation of the *Pinacoteca Nazionale di Bologna* as a cultural context: its collection, its physical spaces, the type of artworks displayed, and the possible difficulties that young or non-specialist visitors may experience during a traditional museum visit.

From this initial analysis, the design process moved toward the definition of a specific user experience problem: how to make young visitors more active, attentive, and emotionally involved in front of artworks that may otherwise feel distant or difficult to interpret. To address this issue, personas were created and the PACT framework was also used.

CoDesign helped transform the initial problem into a game concept. Through brainstorming, comparison with existing references, and progressive refinement, the project evolved into a hybrid cooperative board game with a companion app. The design process focused on connecting physical components, such as the board and cards, with digital interaction, narrative progression, and museum exploration.

The concept was then developed through several design outputs: the game loop, game mechanics, user flowchart, swimlane flowchart, user journey map, and interactive narrative with Twine. These tools made it possible to move from a general idea to a structured experience, defining how players start the game, interact with artworks, complete challenges, receive feedback, and progress toward the final goal.

| Design activity    | Output produced                                                                                                                                                                         | Where it appears in this paper |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Concept design     | Definition of A Very Holy Mess as a cooperative hybrid game for the <i>Pinacoteca Nazionale di Bologna</i> , aimed at making young visitors interact more actively with the collection. | Section 4                      |
| Game loop          | Definition of the repeated cycle of play: receive a quest, move through a zone, interact with an artwork, complete a challenge, and progress in the story.                              | Section 5                      |
| Game mechanics     | Definition of the main mechanics: Artwork Cards, Challenge Cards, board zones, timer, QR/AR interaction, mini-games, and win/lose conditions.                                           | Section 5                      |
| User flowchart     | Step-by-step representation of the user's interaction with the app, from onboarding to mission progression and final outcome.                                                           | Section 6                      |
| Swimlane flowchart | Mapping of the interaction between the player, physical components, companion app, artworks, and game system.                                                                           | Section 6                      |
| User journey map   | Description of the experience over time, including actions, emotions, expectations, pain points, and design responses. Narrative prototype showing the story progression with Twine.    | Section 7                      |

### 3.2 Data/materials used: cards, museum sources, student observations, references

The project was supported by a combination of museum research, collection analysis, student observations, and benchmark references. The first research activity focused on the *Pinacoteca Nazionale di Bologna* as both a cultural institution and a physical exhibition space. We analysed the museum's collection, the distribution of the artworks across the rooms, and the way visitors are guided through the exhibition path. This helped identify which paintings could become meaningful points of interaction within the game.

Particular attention was given to the artworks selected for the narrative path. Vitale da Bologna's *San Giorgio e il drago* was studied as the main narrative trigger, while the other paintings were selected according to their visual richness, symbolic content, position in the museum route, and potential to support challenges based on observation, detail recognition, and interpretation. The artworks were therefore not treated only as visual assets, but as sources for narrative, gameplay, and educational content.

The research also included observation of the museum’s communication and visitor experience. We considered the limits of a traditional visit, the possible difficulty of approaching religious and historical paintings without mediation, and the lack of strong digital continuity after the visit. These observations informed the choice to design an experience that could work both inside the museum and at home.

Benchmark research was another important part of the process. *Murder Party Pocket* and *Chronicles of Crime* were analysed as references for hybrid board game mechanics, companion app guidance, physical cards, and QR-based interaction. The Museum Mystery and Mostri agli Uffizi were used as museum-related references for playful exploration, clues, and hidden details. This comparison helped define the project gap: combining the replayability and structure of hybrid board games with direct engagement with real museum artworks.

## 4. Design Brief

Describe the design Brief [use the design-brief summary in attach as reference]

| Design brief element                                                              |                                                                                                                                                                                                                                                                                                                                                                              | Student answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title                                                                             |                                                                                                                                                                                                                                                                                                                                                                              | A Very Holy Mess                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Type of application / game                                                        |                                                                                                                                                                                                                                                                                                                                                                              | Cooperative hybrid board game with a companion app                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Primary audience                                                                  |                                                                                                                                                                                                                                                                                                                                                                              | Middle and high school students                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Secondary audience                                                                |                                                                                                                                                                                                                                                                                                                                                                              | Young adults                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Primary persona, short description                                                |                                                                                                                                                                                                                                                                                                                                                                              | Gabriel is a restless 13-year-old middle school student who despises sitting still in a classroom and prefers to be physically active. He doesn't know the <i>Pinacoteca Nazionale di Bologna</i> and does not care much for art. He would prefer anything but listen to the professors. He is confident using smartphones and digital content, and he is more likely to engage when learning feels informal, playful, and different from a normal classroom activity. He's most motivated by any kind of atypical lesson format.                                                                                                                             |
| Secondary persona, short description                                              |                                                                                                                                                                                                                                                                                                                                                                              | Maria is a 20-year-old tabletop game enthusiast who is always looking for new games and hybrid experiences. She is sceptical of educational games, because they often feel less immersive and do not always make learning genuinely fun. Maria uses a wheelchair, and although she is interested in art and cultural experiences, she often feels excluded from museum activities because many of them depend on physical presence, movement through the exhibition space, or spaces that are not fully accessible. For this reason, she is motivated by experiences that allow her to participate without feeling limited by the museum's physical barriers. |
| Project focus / context                                                           |                                                                                                                                                                                                                                                                                                                                                                              | Expanding and strengthening the communicative and educational potential of the <i>Pinacoteca Nazionale di Bologna</i> 's collection, using Vitale da Bologna's <i>San Giorgio e il drago</i> as a narrative entry point to support a more active, accessible, and engaging relationship between young visitors and the artworks.                                                                                                                                                                                                                                                                                                                              |
| Star asset                                                                        | Short description                                                                                                                                                                                                                                                                                                                                                            | Media / image placeholder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| San Giorgio e il drago by Vitale da Bologna                                       | The painting represents one of the most recognizable conflicts in Christian iconography: Saint George facing the dragon. It's also the narrative starting point of the project, becoming the emotional and symbolic core of the experience: the player is asked to help Saint George, transforming the artwork from a static object into the beginning of an active mission. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Artworks by major Italian painters, such as Giotto, Carracci, Perugino and others | A curated selection of the museum's core collection that serves as the interactive checkpoints for the game's levels. By transforming these static historical artworks into dynamic puzzle hubs, the game forces players to slow down, look closer, and actively interface with different art eras.                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

Polittico by Giotto

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|                                 |                                                                                                                                                                                                                |                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                        | Pala Scarani by Perugino                                                                                                                                                                            |  |
| <b>Institution / client</b>     |                                                                                                                                                                                                                | <b>Institutional goals</b>                                                                                                                                                                          |                                                                                                                                                                                                                                                                                        | <b>Audience goals</b>                                                                                                                                                                               |  |
| Pinacoteca Nazionale di Bologna |                                                                                                                                                                                                                | 1. Increase visitor participation: make the museum visit more active and engaging.<br><br>2. Attract new demographics: reach young visitors, students, families, and non-specialist audiences.      |                                                                                                                                                                                                                                                                                        | 1. Foster cognitive processes: stimulate observation, memory, interpretation, and problem-solving.<br><br>2. Improve group familiarity: encourage cooperation, communication, and shared decisions. |  |
| <b>Audience</b>                 | <b>Motivations</b>                                                                                                                                                                                             | <b>Barriers</b>                                                                                                                                                                                     | <b>Capabilities</b>                                                                                                                                                                                                                                                                    | <b>Devices</b>                                                                                                                                                                                      |  |
| Primary audience                | 1. Engaging in a dynamic, mission-driven activity (a secret quest) rather than passive observation.<br><br>2. Using familiar technology (smartphones/AR) to interact playfully with the environment.           | 1. Short attention spans and quick loss of interest in traditional, static museum environments.<br><br>2. Lack of prior knowledge or spontaneous interest in religious and historical art.          | 1. High digital literacy and intuitive use of mobile interfaces.<br><br>2. Quick puzzle-solving and visual scanning skills.<br><br>3. Ability to cooperate and communicate actively within a peer group.                                                                               | 1. Personal smartphone.<br><br>2. Companion App.<br><br>3. Physical game components.                                                                                                                |  |
| Secondary audience              | 1. Playing a structurally sound, challenging, and replayable hybrid board game.<br><br>2. Finding inclusive cultural experiences that overcome the physical accessibility limits of traditional museum spaces. | 1. Physical barriers (e.g., wheelchair navigation fatigue or inaccessible rooms) inside the museum.<br><br>2. Scepticism toward "edutainment", which often feels superficial and not genuinely fun. | 1. Advanced understanding of tabletop game mechanics, rules, and hybrid digital/physical integration.<br><br>2. High critical thinking, strategic planning, and observation skills.<br><br>3. High cultural sensitivity and willingness to deeply analyze details if properly engaged. | 1. Personal smartphone.<br><br>2. Companion App.<br><br>3. Physical game components.                                                                                                                |  |

## 5. Design: Concept and Mechanics

*Explain how the project works. In case of serious game, distinguish the game loop from the mechanics.*

### 5.1 Main idea of the project / user experience description

*A Very Holy Mess* is a hybrid phygital adventure designed as an immersive extension of the *Pinacoteca Nazionale di Bologna*. The project connects the physical museum visit with digital storytelling, transforming the artworks into active parts of a narrative game. The player takes on the role of Saint George's guardian angel and must help him defeat the dragon after he loses his weapon.

The experience begins in front of Vitale da Bologna's *San Giorgio e il drago*, where Saint George asks the player for help. From that moment, the museum visit becomes a quest: the player moves through different sections of the Pinacoteca, interacts with selected artworks through the companion app, receives clues from painted characters, and completes mini-challenges connected to visual observation and interpretation.

The main game loop is based on a repeated sequence: the player receives a mission, moves to a specific artwork or museum zone, frames the painting with the app, listens to the character's instructions, completes a mini-game, and unlocks the next step of the story. Each challenge requires the player to look carefully at the artwork and perform simple gesture-based actions, such as tracing an outline, pressing and holding, tapping, scrolling, or identifying hidden details. To avoid technical barriers, we have prioritized intuitive, gesture-based controls that do not require complex tech knowledge, ensuring that the "deep looking" experience is never derailed by the interface. Success is determined by the player's ability to maintain focus and execute interactions accurately, with the quest culminating when the user finally returns the sword to Saint George, allowing him to strike the dragon and succeed.

The app is not used as a simple audio guide, but as a reactive narrative tool. It gives instructions, validates the player's actions, manages progression, and connects each success or failure to the story. This allows the digital layer to support the museum experience without replacing the physical encounter with the artworks.

The game also includes an at-home version. In this mode, the museum route is recreated through a board, Artwork Cards and Challenge Cards. The cards reproduce the selected paintings and act as physical keys: by scanning them, players unlock the same digital dialogues, clues, and mini-games through the app. In this way, the experience can continue outside the museum while maintaining a strong connection to the original collection.

### 5.2 Winning/completion condition and scoring system

The game is completed when the players successfully recover and empower Saint George's weapon, return to *San Giorgio e il drago*, and allow him to defeat the dragon. To reach this ending, players must complete all the required quests across the museum zones.

The main win condition is therefore narrative and time-based rather than based on a scoring system. Players win if they complete the full quest before the main timer runs out. If time expires, the dragon reaches Saint George and the game ends in defeat. A special early game-over condition appears during the hiding challenge: if players choose the wrong painting as a refuge for Saint George, he is found by the dragon and the mission fails.

The companion app handles the unlocking of new narrative steps for each completed challenge and marks progress toward the final quest. Some mini-games include local scoring objectives, such as reaching a required number of points, finding all requested figures, or completing a precise gesture within the time limit. Other challenges work through success/failure validation: if the player makes a mistake, the app asks them to retry, which makes them lose time on the challenge.

In the at-home board game version, time also functions as a balancing mechanic. Since players are not physically moving through the museum, crossing from one zone of the board to another subtracts time from the main timer, simulating the travel time of the museum visit. This keeps both versions of the game connected to the same tension: players must observe carefully, but also act quickly enough to save Saint George.

## 6. User Flow, Interaction and Hybrid System

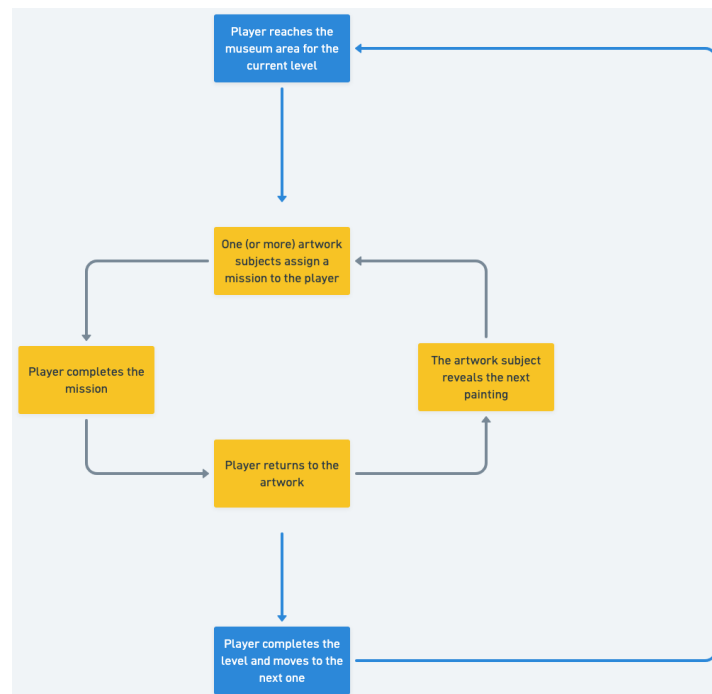
Insert diagrams here. Each figure must have a caption and must be discussed in the text.

### 6.1 General user flow description

The basic user dynamic revolves around completing a given number of small challenges and advancing accordingly to the next level. More specifically, there are two distinct loops. The first one (the blue loop in the figure) describes the overall game progression, requiring the player to move through the museum spaces to reach the area corresponding to the current level, which they cannot leave until the level is completed.

The second one (the yellow loop in the figure) involves the inner dynamic of each level: an artwork subject assigns the player a small challenge, which can be solved by carefully observing and deducing clues from other artworks (either physically or through cards, depending on the player's location). Once the player finds the solution, they return to the original artwork and use the app to validate their answer. If correct, the artwork subject reveals the next painting, triggering the next challenge within the level loop until all conditions to advance are met.

**Figure 1. General user flow diagram / flowchart**



### 6.2 Swimlane flowchart description: actors, responsibilities, decisions and failure states

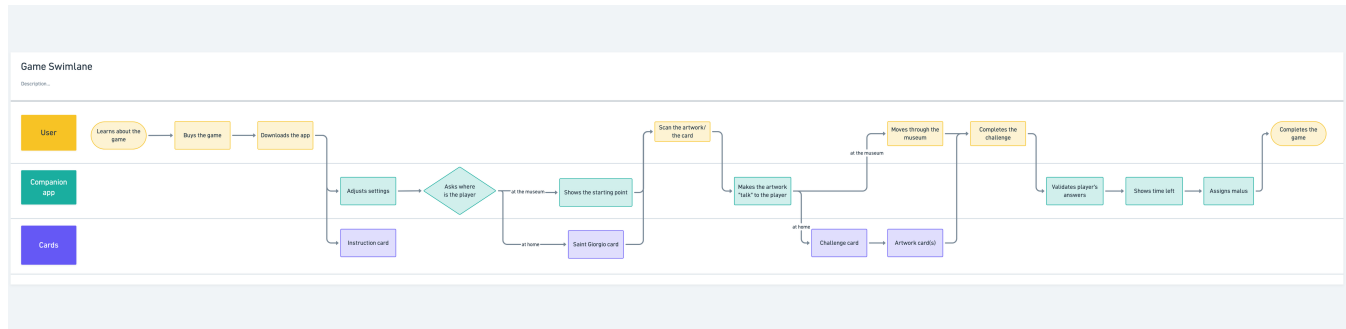
The game dynamic is based on the interaction between three main actors: the player, the companion app, and the card decks.

The player interacts with the app, follows the instructions provided, and scans the artworks to discover the challenges they must solve to unlock the next level. To complete the mission and defeat the dragon, the player physically moves through the museum, observing the artworks and using their critical thinking to solve the challenges. If the player is playing at home, they draw cards from both the challenge deck and the artwork deck.

The companion app is needed whether the player is at the museum or not. In both cases, the app allows the player to adjust settings (such as volume, language, accessibility options, etc.), scan and interact with the artworks, receive new challenges, and validate their answers. The app also keeps track of the time left and the current level.

At the beginning of the game, the app asks the player where they are, because some of the aforementioned functions are performed slightly differently depending on the player's location. At the museum, the app animates the paintings when the player scans the physical artwork at the Pinacoteca, which assigns a challenge to the player. If the player is at home, the app animates the artwork cards, allows the player to see an HD version of the image, and prompts the player to draw a card from the challenge deck.

The decks are used for the at-home version. The Instruction card, together with the app, helps the player understand how to set up the game and use the decks. The artwork cards replace the physical artworks: they can be scanned to discover challenges, talk to the paintings, and analyse a high-quality version of the artwork. Finally, the special "Saint George" card explains the background story and asks for the player's help, while the challenge cards assign the player new random challenges to solve in order to advance through the levels.

**Figure 2. Swimlane flowchart**

### 6.3 Interaction description and interaction categories

The game is designed as a physical-digital hybrid experience. The player uses with the app to understand the game rules and dynamics, check the time left, validate their answers, awaken the artworks' subjects and interact with them. The physical component is represented by the actual artworks (if the player is at the Pinacoteca), which the player has to carefully observe to solve the challenges; at home, the original paintings (Saint George's one included) are replaced by a physical card deck, and the challenges are drawn from the challenges deck.

The interaction categories can be summarized as follows:

- **Digital-to-Physical (and vice versa):** Scanning physical artworks or cards via the companion app to trigger digital events, such as animating the painting's subject or unlocking a new quest.
- **Player-to-Physical:** Physically exploring the museum space, observing fine details in the paintings to uncover hidden clues, or physically handling the artwork and challenge decks at home.
- **Player-to-Digital:** Interacting with the mobile UI to read instructions, adjust settings, input answers for validation, and communicate with the awakened artwork subjects.

### 6.4 Hybrid and technological layers

The technological layer is represented by the companion app, which creates a more dynamic game experience by “awakening” otherwise static artworks, providing feedback, and delivering enjoyable educational content. The QR codes on the cards, the image-recognition software, and augmented reality (AR) build the bridge between the physical and technological layers, allowing the player to interact with the paintings and learn in an innovative, dynamic way.

| Layer          | Software / hardware / material              | Function                                                                        |
|----------------|---------------------------------------------|---------------------------------------------------------------------------------|
| Physical layer | Cards and board (only for at-home version)  | Obtain educational information, observe and analyse the artwork                 |
| Digital layer  | Companion app                               | Set up the game, animate artworks, keep time, apply penalties, validate answers |
| Museum layer   | Artworks                                    | Obtain educational information, observe and analyse the artwork                 |
| Social layer   | Interaction and cooperation between players | Solve challenges more easily, promote teamwork                                  |

## 7. User Experience as User Journey

Describe the experience over time from the user's point of view: phases, actions, emotions, touchpoints, pain points, opportunities. Expand the storyboard. Add the Twine.

### 7.1 User journey narrative

The user journey begins with a sense of initial scepticism when **Gabriel**, a 13yearsold student, arrives at the *Pinacoteca Nazionale di Bologna* with his class. He doesn't enjoy going to school every day, since the idea of being stuck in a chair for hours makes him uncomfortable. Gabriel expects to attend a traditional visit in a museum, where the guide shows a painting and talks about it for hours. However, the expectation of a passive museum visit is quickly subverted when the user, in this case Gabriel and his class, see the manifesto with the advertisement of the app. And when they meet Saint George immediately the attitude change from a passive observation to an active participation. Gabriel, who was firstly not interested in the visit, now feels the urgency to help the Saint. To do that, however, the Saint himself tells him that he needs his lost sword. Gabriel starts to play to finally find that arm:

**Level 1:** initially he is directed to the medieval section, where he attends some challenges from Archangel Michael in *Giotto's Polyptych* and learns about the punishments of hell in *Heaven and Hell*

**Level 2:** then he enters in the Renaissance section and so pass from the gold medieval backgrounds to skies full of angels, as he sees first in the *Pala Manzuoli* and then in the *Pala Scaranì*, which gives him more challenges. Then Gabriel finally finds the sword in *Madonna col Bambino in gloria e i Santi Michele Arcangelo, Pietro e Benedetto*.

**Level 3:** After a little break, in which he needs to hide in one of the paintings of the level 2, Gabriel, guided by the Gozzadini's family, ends up in the Carracci's area, in which he must find the details, proving that an artist can convey a great deal even through them.

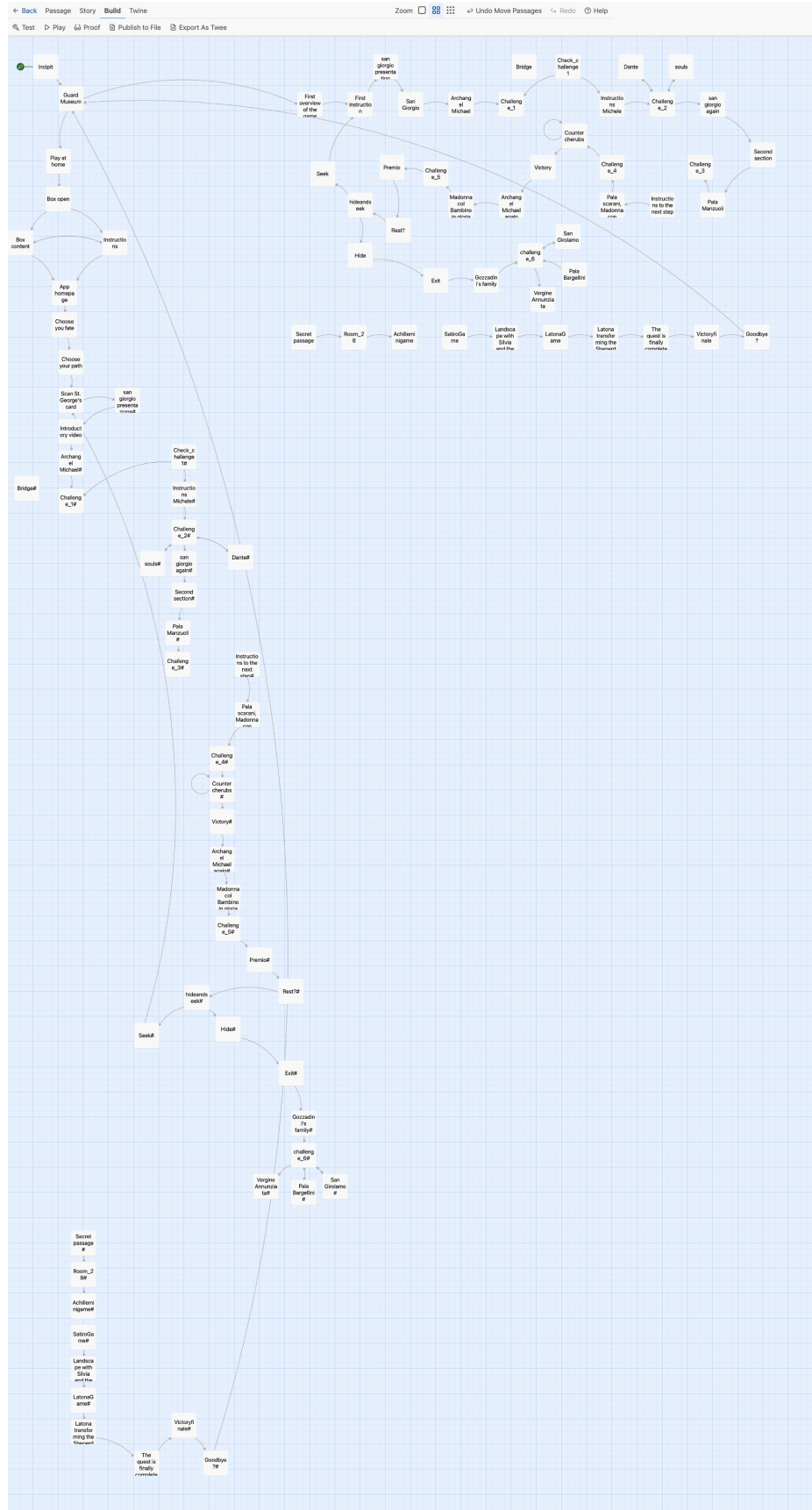
**Level 4:** Gabriel realizes that to complete the game and make the sword stronger enough to kill the dragon, he must reach the end of the museum's path; he thus arrives at the final zone, where the landscapes begin to take on Neoclassical lines, transforming into majestic, pagan vistas.

In the end of level 4 Latona, one of the protagonists of the latest painting, helps him and give the sword the fierce needed. Gabriel returns from San Giorgio with the sword, and they finally succeed in their quest: the dragon is rendered powerless. Most of all Gabriel emerges victorious, enjoying the visit—appreciating the importance of art without losing sight of the element of fun.

The various interaction in the challenges requires precise manual gestures—such as tracing outlines, scrolling, pressing and holding your finger on the screen—which serve as the core gameplay mechanics; these actions must be repeated if the user fails to succeed in the challenges. The journey is punctuated by moments of intense concentration, particularly when the user must maintain focus to avoid "demonic" penalties. The successful completion of these challenges provides the user with a profound sense of accomplishment and narrative progression. Ultimately, the user emerges from the experience with a deeper, active connection to the historical masterpieces and a solid knowledge of the various paintings with which he plays. In the end the goal, in fact, is not just to entertain and make the visit more enjoyable, but also to educate and to enrich the user's artistic knowledge through an educational, hybrid framework.

To make the experience truly hybrid and ensure it continues well beyond the museum's walls, we have also designed a tabletop version of the game, complete with a game board and a set of cards. We have developed a dedicated Twine path for this version as well, which mirrors the original museum experience but has been reimagined so it can be played anywhere. It remains a deeply hybrid concept: while the museum visit revolves around scanning the actual artworks with the smartphone, this home version adapts the mechanics by using the physical artwork cards, keeping that same connection between gameplay and art even when you're at home.

Figure 3. User Journey Map / UX diagram



| Phase                    | User action                                                                                                                        | Touchpoint                                                                                              | Emotion / expectation                                                                                                      | Pain point                                                                     | Design response                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before visit / discovery | Going to the museum                                                                                                                | Ad manifesto at the museum/<br>Buy the Game                                                             | Skeptical, bored                                                                                                           | The traditional visit feels like a passive school obligation                   | The narrative hook transforms the visit into a more enjoyable secret mission.                                                                                                    |
| Onboarding               | Downloading and setting the app                                                                                                    | Smartphone interface with the app                                                                       | Curious                                                                                                                    | Traditional setup process                                                      | Immersive voice of the protagonist (Saint George)                                                                                                                                |
| Mission / Goals          | Framing the painting using the app, and one of the characters within it will illustrate the mission/<br><br>Use the artworks cards | AR interaction / QR code                                                                                | Engaged, determined, active in participation, focus during the challenges                                                  | Ambiguity in mission goals                                                     | Narrative-driven instruction. The paintings' characters are clear when they ask the user what is needed and the various directives for the challenges                            |
| Capture / interaction    | Performing precise gestures (tracing, tapping, pressing, scrolling)                                                                | Screen interaction paintings                                                                            | Feeling of adrenaline, nervous, afraid of making a mistake                                                                 | Fear of making mistakes during manual interactions                             | The interaction with the screen are easy and manageable                                                                                                                          |
| Feedback / scoring       | Complete the mini-challenges to obtain the final quest                                                                             | Narrative progress. If you win the challenge, then you can go on, otherwise you have to retry the game. | Fulfilled, accomplished, confident, felt valued, and contributed to the cause.                                             | Difficulty in correlating digital feedback with the physical artwork's meaning | The app, thanks to the characters' paintings, provides narrative-driven explanations to reinforce learning.                                                                      |
| After experience         | Reflexive interaction with artworks                                                                                                | Solid knowledge about some museum's painting                                                            | Satisfied not to have sat through the usual kind of visit, but to have managed to gain new insights while also having fun. | Difficulty leaving the digital world                                           | As it is a hybrid game, where you also had to engage with the actual work (or the cards if you are playing at home), there is nonetheless a strong connection to the real world. |

### Interactive story / Twine link

[https://github.com/elemocc/A\\_Very\\_Holy\\_Mess](https://github.com/elemocc/A_Very_Holy_Mess)

## 8. Response to Pain Points and Innovation

| Pain point identified in Section 1                                                      | Design solution                                                                                                                                                                                                                                                                                                                                    | Expected effect                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The museum contains many artworks, which can make the visit feel visually overwhelming. | The museum is divided into four zones, each connected to a specific quest, difficulty level, and set of challenges. Players are not asked to explore everything at once, but to move through structured areas with clear objectives. At home, the four zones are translated into the board and a different deck of Challenge Cards for every zone. | The visit becomes easier to navigate and less cognitively overwhelming. Players understand where to go, what to focus on, and why each artwork matters within the game progression. |
| Visitors may look quickly and miss important details.                                   | The game requires players to interact with artworks in order to solve quests, ask the right questions, and complete mini-game challenges based on visual clues, symbols, gestures, characters, and narrative elements.                                                                                                                             | Observation becomes slower, more careful, and goal oriented. Players are encouraged to analyse the artworks actively rather than simply looking at them briefly.                    |
| Religious and historical subjects may feel distant from young audiences.                | The project uses Saint George and the dragon as a humorous and emotionally accessible narrative framework. Players become Saint                                                                                                                                                                                                                    | The artworks become part of a living story rather than distant historical objects. Young players can connect emotionally with the                                                   |

|                                                                      |                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                      | George's guardian angel and must help him recover what he needs to defeat the dragon.                                                                                                                                                                                                                                                                                 | protagonist and understand religious or historical images through action, dialogue, and play.                                                                                                                                              |
| The museum has limited digital extension outside the physical visit. | The museum experience continues beyond the physical visit through the boardgame. The companion app allows the experience to exist both inside and outside the museum: in the museum, the app works as an AR tool that brings paintings to life; at home, players can scan QR codes on Artwork Cards to interact with reconstructed digital versions of the paintings. | The museum experience continues beyond the physical visit through the boardgame and the companion app. The project supports remote engagement, post-visit play, and a stronger connection between the museum collection and everyday life. |
| The visit often lacks a reason to be repeated.                       | The game includes randomised challenges (both in the museum through the app and at home through the Challenge Cards), different possibilities of interactions with the artworks, and a time limit. Even if the main quests remain fixed, the challenges and the path through the experience can change.                                                               | The experience gains re-playability. Visitors have a reason to return to the museum, replay the game with different people, or continue engaging with the artworks from home.                                                              |

### 8.1 Innovation beyond the state of the art

*A Very Holy Mess* introduces several specific innovations that go beyond the existing hybrid board games and museum games researched before end.

The first innovation lies in transforming artwork observation into an active, social, and memorable learning-by-doing experience. Instead of being a simple museum guide, through gamification players do not passively receive information about the artworks of the Pinacoteca, but naturally build understanding through quests, dialogue, cooperation, and visual investigation.

Another innovation is the application of hybrid investigation mechanics to real artworks. Games such as *Murder Party Pocket* and *Chronicles of Crime* use physical cards, apps, and QR codes to support fictional mystery stories. *A Very Holy Mess* transfers this logic into a museum context, using the same type of phygital interaction to make players observe paintings, identify details, ask questions, and interpret narrative or symbolic elements.

A further innovative aspect is the double play model. The game is designed to work both inside the museum and at home. In the museum, players interact with the real artworks through the companion app; at home, they use Artwork Cards and QR codes to recreate the encounter with the paintings. This extends the museum experience beyond the physical visit and transforms the at-home version into a continuation of cultural engagement, not only a simplified substitute.

Another important innovation is the reusability of the physical components. The same board, Artwork Cards, and Challenge Cards can support multiple play sessions, because the main quests remain stable while the challenges can be randomised. This makes the game replayable, and capable of generating repeated engagement with the museum collection over time.

Finally, the game introduces a specific emotional role for the player towards the real artworks. This turns the visitor from an external observer into an active helper inside the narrative. In this way, the artwork becomes not only something to look at, but the starting point of a cooperative mission.

*Explain what is new: a specific hybrid mechanism, a new use of museum assets, an original combination of photography learning and heritage interpretation, a new collaborative structure, a new evaluation/validation method, etc.*

## 9. Conclusion

*Summarise the project, contribution, limitations and next steps in 150–250 words.*

### Conclusion text

*A Very Holy Mess* responds to a concrete set of problems identified in relation to the *Pinacoteca Nazionale di Bologna*: the risk of visual overload caused by a large and complex collection, the tendency of visitors to observe artworks quickly and superficially, the distance that religious and historical subjects can create for young audiences, the limited digital extension of the museum experience, and the lack of strong reasons to repeat the visit. The project addresses these issues by transforming the museum path into an active, playful, and cooperative mission, where the player becomes Saint George's guardian angel and is asked to save him through observation, interpretation, and interaction with the artworks.

Through a human-centred design process based on museum analysis, benchmarking, persona creation, co-design, the PACT framework, user flow mapping, swimlane diagrams, user journey design, and Twine prototyping, we developed a phygital hybrid game that connects physical artworks, physical cards, and a companion app. The main contribution of the project is the use of hybrid board game mechanics not for a fictional mystery, but to create a learning-by-doing experience around real museum artworks. The game encourages players to slow down, cooperate, notice visual details, and build knowledge through narrative progression and mini-games.

From a design perspective, we are proud of having created a concept that can work both inside the museum and at home, extending the cultural experience beyond the visit. However, the project has not yet reached a complete MVP and has not yet been formally tested with final users. The next steps would include usability testing with students, families, and young adults, refinement of the mini-games, accessibility improvements, and the technical development of the AR and QR-based interactions.

## Acknowledgments and Team Roles

| Team member       | Role in project design                                                                               | Role in writing the paper |
|-------------------|------------------------------------------------------------------------------------------------------|---------------------------|
| Chiara Genovese   | Visual designer, Prototype designer / technical lead                                                 | Section 1, 2, 4, 5, 8, 9  |
| Kielyn C. Madayag | Narrative / Concepts designer, Experience and Interaction Designer (Twine for the museum experience) | Section 1, 2, 5, 7, 9     |
| Elena Mocci       | Research / content lead, Experience and Interaction Designer (Twine for the at home experience)      | Section 1, 2, 3, 4, 6, 9  |

## AI statement

The authors used AI-assisted tools to support the writing and generation of images and videos as a sample for what the final product would look like. The AI tools used are: ChatGPT by OpenAI and Gemini by Google for image processing, and Pippit by ByteDance for video generation. All content, design decisions, and research findings are the result of the authors' own work. The AI was used as a writing/graphic support tool and not as a source of data or design decisions.

## References

Add publications in alphabetical order. Use a consistent citation style. Include all sources cited in the text and figure credits.

[Author Surname], [Initials]. ([Year]). [Title]. [Journal/Book/Publisher]. [DOI/URL if available].

Pescarin, S., Massidda, M., Travaglini, L., Spotti, S., & Veggi, M. (2025). PERCEIVE Co-design Tool for Cultural Heritage Experiences. Zenodo. <https://doi.org/10.5281/zenodo.15173219>

Spotti, S., Bonifazi, F., Massidda, M., Pescarin, S., & Travaglini, L. (2025). Designing Authentic Digital Experiences. Digital Heritage 2025.

## Web Links

### Web links used for benchmarks, museum sources, tools, datasets, images

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## Appendix: Design Brief Summary

Our Project, *A Very Holy Mess* is a cooperative hybrid board game with a companion app designed for the *Pinacoteca Nazionale di Bologna*.

Its primary audience is middle and high school students, and its secondary audience is young adults.

Their characteristics are as follows. Our primary persona is Gabriel, a restless 13-year-old student who dislikes passive school activities, has little spontaneous interest in art, and is more likely to engage when learning feels active, playful, and mediated through familiar digital tools. Our secondary persona is Maria, a 20-year-old tabletop game enthusiast who is interested in cultural experiences but needs them to be accessible, re-playable, and engaging also outside the physical museum.

The context of our project is the expansion and strengthening of the communicative and educational potential of the *Pinacoteca Nazionale di Bologna*'s collection. The project uses *San Giorgio e il drago* as a narrative entry point to create a more active, accessible, and emotionally engaging relationship between young visitors and the artworks.

The main and general pain points are: the visual overload caused by a large and complex collection, the tendency of visitors to observe artworks quickly and superficially, the distance that religious and historical subjects may create for young audiences, the limited digital extension of the museum experience, and the lack of strong reasons to repeat the visit.

Analysing what has been already produced and considering what is already available as a solution I have found good references in *Murder Party Pocket* and *Chronicles of Crime*, which combine physical cards with app-guided progression and QR-based interaction. We also considered *The Museum Mystery* and *Mostri agli Uffizi* as references for playful museum exploration, clues, hidden details, and observation-based activities.

Nevertheless the solutions already available do not solve the following gaps: hybrid investigation board games are usually not made to be educational, while museum games often remain closer to guided activities or scavenger hunts and do not offer the same re-playability and phygital structure of modern hybrid board games.

Star Assets of the project are *San Giorgio e il drago* by Vitale da Bologna and the curated selection of artworks by major Italian painters such as Giotto, Perugino, Carracci, Lavinia Fontana, and others.

The reference institution is the *Pinacoteca Nazionale di Bologna*, and its institutional goals are to increase visitor participation by making the museum visit more active and engaging, and to attract new demographics such as young visitors, students, families, and non-specialist audiences.

The audience goals our project is targeting are fostering cognitive processes, such as observation, memory, interpretation, and problem-solving, and improving group familiarity through cooperation, communication, and shared decision-making.

Our primary target audience would be motivated to experience my project by the possibility of taking part in a dynamic mission instead of a passive museum visit, and by the use of familiar technology such as smartphones and AR interaction; it might experience specific barriers like short attention spans, low interest in traditional museum formats, and limited prior knowledge of religious or historical art. The Audience is nevertheless capable of using mobile interfaces, solving visual puzzles, cooperating with peers, and it can be able to use devices such as personal smartphones, the companion app, and physical game components, and these have been used as baseline for the design.

My secondary target audience would be motivated to experience my project by the possibility of playing a structurally solid and re-playable hybrid board game, and by the opportunity to access a cultural experience also from home; it might experience specific barriers like physical accessibility issues inside the museum and scepticism toward educational games that feel superficial. The Audience is nevertheless capable of understanding board game mechanics, using hybrid phygital systems, analysing visual details, and it can be able to use devices such as personal smartphones, the companion app, and physical game components, and these have been used as baseline for the design.

Main Idea of the project is to transform the museum visit into a cooperative quest. The player starts from *San Giorgio e il drago*, receives a mission from Saint George, moves through the museum zones, frames selected artworks with the app, listens to animated characters, completes mini-games, and progresses through the story until the weapon is recovered and empowered. The general user flow is represented in the flowchart in Section 6.

To better describe our project, this is the User Journey. The user journey follows Gabriel from an initial state of boredom and scepticism to active involvement. At first, the museum visit appears to him as a passive school activity, but the app transforms the visit into a secret mission. Gabriel meets Saint George, receives the role of guardian angel, explores the museum sections, completes challenges, learns to observe details, and finally returns to Saint George with the empowered weapon. The experience is accessible also as interactive story in [https://github.com/elemocc/A\\_Very\\_Holy\\_Mess](https://github.com/elemocc/A_Very_Holy_Mess).

The interaction includes both physical and digital actions. Players scan artworks or Artwork Cards, listen to app-based dialogues, observe painting details, perform gesture-based mini-games, validate answers, and cooperate with other players. The main interaction categories are digital-to-physical interaction, player-to-physical interaction, and player-to-digital interaction.

The hybrid elements and technological elements and their role include the companion app, AR interaction with the real artworks, QR-code scanning for Artwork Cards, the game board, physical cards, and the museum artworks themselves. The app manages onboarding, narrative progression, feedback, timer, penalties, challenge validation, and digital animation. The physical components support re-playability and allow the game to continue at home.

The project addresses the above-mentioned pain-points by dividing the museum into clear zones, giving players specific objectives, requiring careful observation of artworks, using Saint George as an accessible narrative frame, extending the experience through an at-home version, and supporting re-playability through physical cards, randomised challenges, and timed progression.

It's innovation lays in applying hybrid board game mechanics to real museum artworks. Instead of using physical cards, app interaction, and QR codes to solve a fictional mystery, *A Very Holy Mess* uses them to make players observe, question, interpret, and emotionally connect with cultural heritage. The project also introduces a double play model, allowing the experience to function both inside the museum and at home, and transforms the visitor from a passive observer into an active helper inside the narrative.